SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.* (BTL 3)

Assessment Tool Weightage: 40%

QUESTIONS

2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.

3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

11. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications.

17. An image captured under uneven lighting shows poor contrast. Explain how the image formation process contributes to this issue and suggest corrective measures.

22. A digital camera produces images with banding effects. Analyze how quantization levels contribute to this issue and propose a solution.

33. A low-bit image loses important details. Explain how quantization affects information preservation.

40. A digital image shows quantization noise. Explain its origin and suggest mitigation techniques.

43. In a facial recognition system, explain how resolution impacts feature extraction accuracy.

46. A system requires detecting small objects in images. Explain how sampling and resolution should be adjusted.

2 Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.

1 Real-world scene (low light)



- Light levels are low, so less light reaches the camera.
- Scene appears dark with poor details to the human eye.

2 Lens collects the light



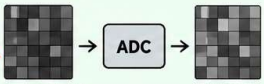
- The lens gathers as much light as possible and focuses it onto the image sensor.
- In low light, the lens aperture stays more open (larger f-number like f/1.8) to let in more light.

3 Image sensor captures light



- The sensor (CCD/CMOS) converts incoming light into electrical charge.
- In low light, the sensor needs a longer exposure time and higher gain (ISO) to collect enough light.
- This can introduce more noise and reduce detail.

4 Analog to digital conversion (ADC)



- The electrical signal is converted into digital values (pixels).
- Low light can cause lower signal strength, so the values may be noisy and less accurate.

5 Image processing & noise reduction



- The camera's image processor reduces noise using algorithms (e.g., denoising, demosaicing).
- It also adjusts brightness, contrast and colour based on the low-light conditions.

6 Resulting digital image



- The final image may still have some noise, reduced sharpness and lower detail compared to well-lit scenes.
- Colours may look duller and shadows can lose detail.

7 Key factors that influence the final image

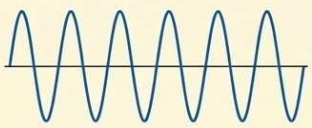
- **Exposure time** (longer exposure = more light)
- **Gain** (higher ISO = brighter but noisier)
- **Aperture size** (larger aperture = more light)
- **Sensor size & quality** (better sensors = less noise)
- **Image processing algorithms** (denoising, enhancement)

8 Conclusion

In low light, the image sensing and acquisition process plays a major role in the final digital image. It controls how much light is collected, how the signal is converted and processed, and how much noise and loss of detail occurs. The result is a darker image with reduced clarity unless proper exposure, ISO and processing techniques are used.

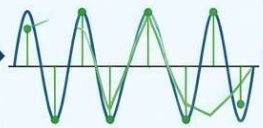
3 3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

1 Original Signal



- This is the original, continuous analog signal (e.g., sound or visual detail).
- Features smooth, rapid oscillations.
- Contains critical high-frequency information.

2 Undersampling (below Nyquist rate)



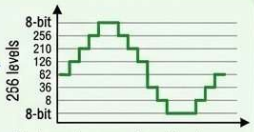
- Sampling rate is too low for the highest frequency.
- < 2x highest frequency (below Nyquist rate).
- The sparse samples cannot reconstruct the fast oscillations.

3 Sampling (discretization in time/space)



- The continuous signal is measured at discrete intervals.
- If intervals are too large, critical information is "skipped".
- This is a fundamental sampling error.

4 Quantization (discretization in amplitude)



- Each analog sample value is assigned to a discrete digital level.
- Limited levels lead to step-like representation (quantization noise/artifacts).
- Less critical for aliasing but affects resolution.

5 Artifact: Jagged edges and Moire patterns



- Because of low sampling, high-frequency details become 'aliased'.
- High frequencies appear as lower-frequency distortions.
- Common manifestations: Jagged diagonals, bizarre moiré patterns in textures.

6 Corrective approach

A. Hardware: Anti-aliasing (Optical Low-Pass) Filter



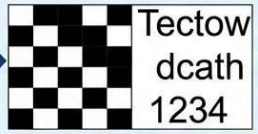
- Apply a physical optical blur filter before the image sensor.
- Blocks high-frequency details (potential aliasing sources) before they are sampled.

B. Software: Increase Sampling (e.g., Super-sampling)



- Capture at a much higher resolution than the target display.
- A dense pixel grid 'over-samples' details, preventing aliasing during capture.

7 Corrected result



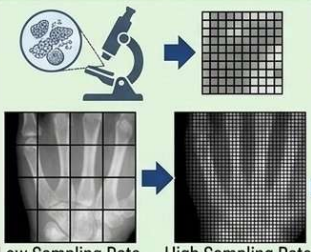
- After applying corrective methods, the image appears sharp and free of visual distortions.
- The actual signal is faithfully represented (or at least looks smooth).

8 Conclusion

- Aliasing is caused primarily by under-sampling, where a signal is sampled below its Nyquist rate, misrepresenting high frequencies.
- The correct approach is either physical filtering (hardware OLPF) to remove the conflicting details or increasing the sampling rate (higher resolution or super-sampling) to capture them correctly.
- Quantization adds noise but is not the primary cause of frequency aliasing artifacts.

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

1 Define Spatial Detail



Low Sampling Rate, 64x64 matrix

High Sampling Rate, 512x512 matrix

I Pixel Resolution (Spatial Resolution)

- Defines the ability to separate small, distinct anatomical structures.
- Governs sharpness and geometric accuracy.

Determined by sampling: mapping the anatomy to a grid of pixels (sampling points).

- Higher matrix (more pixels) gives smaller pixels.

RESULT: Restores high-frequency detail.

II Intensity Resolution (Contrast Resolution)

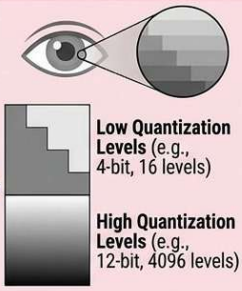
- Defines the ability to distinguish subtle density variations between similar tissue types.
- Governs shades of gray and texture gradients.

Determined by quantization: assigning a specific numerical value (discrete digital level) to each pixel's amplitude.

- Higher bit-depth (more levels) means finer gray gradations.

RESULT: Differentiates subtle contrast.

5 Define Contrast Detail

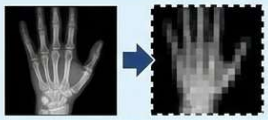


Low Quantization Levels (e.g., 4-bit, 16 levels)

High Quantization Levels (e.g., 12-bit, 4096 levels)

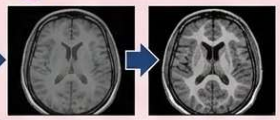
3. Low Spatial Resolution (Few Pixels)

Original X-ray Hamd (like <IMAGE 1>)



- **Blurred image**: small details lost, cannot see micro-fractures, jagged edges.

7. Low Intensity Resolution (Few Levels)



- **Contoured image**: False banding artifacts. Gray and white matter merge into blocky gradients.

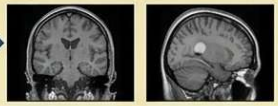
NEWLY CREATED MODULE, to clarify the former confusing panel)

High Spatial Resolution (Many Pixels)



- **Sharp image**: clearly differentiates bone texture, tiny fissures.

8. NEWLY CREATED MODULE, (to balance high res)



- **Smooth image:** Clearly differentiates subtle tissues (e.g., tumor vs. healthy brain).

11. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications. Evaluate sensor performance for a chosen task.

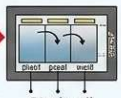
1 Original Scene with High Dynamic Range.



- Real-world, complex scene with high contrast (e.g., strong highlights, deep shadows).
- We want to capture this using CCD and CMOS sensors.

2 Image Acquisition Systems (Sensors)

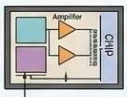
CCD Sensor



Pixel well

Charge-Coupled Device (CCD). Sequential charge transfer, high quantum efficiency.

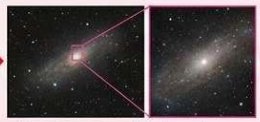
CMOS Sensor



Pixel pixel

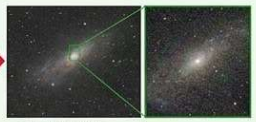
Complementary Metal-Oxide-Semi conductor (CMOS). Amplification at each pixel, random access readout.

3 CCD Sensor Response (High Quality Application)



- Very low noise floor (good for long exposures).
- High uniformity between pixels.
- Large dynamic range.
- Ideal for scientific imaging (e.g., astrophotography).

4 CMOS Sensor Response (High Quality Application)



- Slightly higher noise.
- Poorer uniformity between pixels.
- Lower dynamic range than CCD, but improving rapidly.
- Lower cost, integrated features.
- Less ideal for the *best possible* quality scientific tasks.

5 Effect of Sensor Type on Key Tasks

Parameter	CCD Image Example	CMOS Image Example
1. Dynamic Range	Deep, detailed	Flatter, lost shadows
2. Signal-to-Noise Ratio (SNR)	High SNR, clean	Low SNR, grain
3. Resolution & Noise	Sharp, detailed	Soft, noise-prone

6 Method to Choose Best Sensor (Task-Dependent)



1. For Ultimate Quality & Sensitivity

- Choose CCD sensors for tasks where dynamic range and noise are critical (e.g., astrophotography, microscopy).



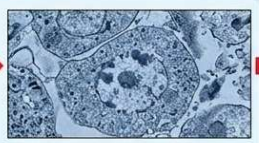
2. For Speed, Integration, & Cost

- Choose CMOS sensors for high-speed capture, smaller size, and consumer devices (e.g., mobile cameras, surveillance).



3. Evaluate Evolving CMOS (New Tech). Assess newer 'stacked' or 'back-illuminated' CMOS. The performance gap with CCD is narrowing significantly.

7 Optimized Task Performance (CCD for Detail)



- CCD sensor is chosen for this demanding task.
- Maximum dynamic range and low noise ensure all critical cell features are clearly visible.
- Superior image quality is essential.

8 Conclusion

- Sensor type (CCD vs CMOS) impacts image quality in practical applications by determining noise, uniformity, and dynamic range. CCD has traditionally offered superior performance for low-light, high-fidelity scientific tasks.
- CMOS offers faster readout, integration, and lower cost for speed-driven consumer applications.
- The gap is rapidly narrowing as CMOS technology improves.
- Choose the sensor based on the prioritized quality parameters for the specific task.

17. An image captured under uneven lighting shows poor contrast. Explain how the image formation process contributes to this issue and suggest corrective measures.

Understand the visual artifacts of improper lighting and how to control them.

1 Original Scene (Uneven Lighting)



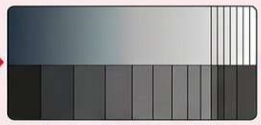
- This is a real-world complex scene with a wide range of continuous light levels (e.g., street lights, shadows). It has high dynamic range (HDR).
- We want to capture this using a digital camera.

2 The Camera Lens & Uneven Exposure



- The lens collects the light from the scene.
- Due to dynamic range limitations, the scene range (light to dark) exceeds the.
- The camera's A/D converter must map continuous scene levels to discrete digital values.

3 Image Sensor Quantization (Origin)



- The sensor (A/D converter) has a limited number of "buckets" (bit depth).
- Continuous levels are rounded (quantized) to the nearest discrete level.
- Shadow details and highlights often exceed the linear range and are quantized into fewer levels or single bins.
- The error (quantization noise) is prominent.

4 The Poor Contrast Image (Result)



- Because of limited sensor dynamic range and poor quantization (few levels for details), smooth gradients are lost.
- Perceptible banding artifacts ("posterization") are visible.
- Small details merge into single colors (blown highlights/crushed shadows).
- Contrast is poor between major blocks.

5 Effect of Dynamic Range on Contrast.

Sensor DR/ Bit Depth	Image Example	Visual Contrast
	**Low** (Few steps/limited range)	**Failure** (Posterization, lost details)
	**Medium** (Standard bit depth)	**Average** (Some banding, adequate details)
	**High** (High bit depth/HDR)	**Smooth** (Clear details, natural gradients)

6 Method to Solve it (Bit Depth & Processing)



1. **Increase Sensor dynamic range/bit depth.**
   - Use a camera with a wider dynamic range and higher bit depth (e.g., from 10-bit to 16-bit capture).



2. **Perform Exposure Bracketing & HDR Merging.**
   - Capture multiple exposures (under, mid, over) and merge them with software for a wider tonal range.
   - More bits → finer steps → seamless gradients.



3. **Use Advanced Post-Processing (Tone Mapping)**
   - Utilize software to remap the captured bits to perceptually smooth gradients and improve contrast, or use deep-learning based restoration.

7 Optimized/HDR Image Continuous Gradients.



- When the dynamic range and bit depth are high, gradients are smooth and natural.
- Detailed information is preserved across the entire range (shadows and highlights).
- Banding artifacts are eliminated.
- Contrast is natural and rich.

8 Conclusion

- **Poor quantization (limited sensor dynamic range and bit depth) causes continuous tonal ranges to be rounded into large discrete blocks, leading to perceptible banding artifacts, especially in deep shadows and harsh highlights.**
- **Smooth gradients are lost through a reduction in tonal fidelity.**
- **Increasing the dynamic range and bit depth is the primary solution, providing more quantization 'buckets' to capture detailed.**
- **A combination of higher bit depth, HDR technique, and post-processing can minimize banding and improve contrast. Finer quantization steps are key to preserving gradients and detail fidelity.**

22. A digital camera produces images with banding effects. Analyze how quantization levels contribute to this issue and propose a solution.

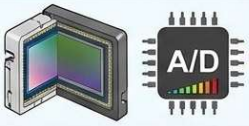
Learn how bit depth and quantization steps affect image gradients and contouring.

1 Original Scene with Smooth Gradients.



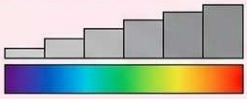
- This is a real-world smooth gradient scene (e.g., sky). It has continuous and seamless tone changes.
- We want to capture this using a digital sensor.

2 The A/D Converter (Digitization).



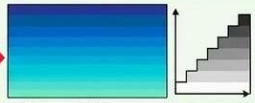
- The A/D converter samples the light and converts analog sensor values into digital units. The bit depth (quantization levels) determines the number of 'steps' available to represent each color.

3 Signal Quantization (Origin).



- Continuous input light values are **rounded (quantized)** to the nearest available discrete digital level.
- A low bit depth (few levels) means large continuous ranges are mapped to single digital 'bins'.
- The difference is quantization error.

4 Image with Banding Artifacts.



- Because of few quantization levels (low bit depth), the image has visible 'banding' or 'posterization'.
- Smooth gradients are lost.
- Subtle tone changes appear as hard, contour-like edges.
- The visual noise is 'stepped'.

5 Effect of Bit Depth on Gradients.

Bit Depth (Levels)	Gradient Example	Banding Quality
Low (e.g., 2 levels)		Severe banding
Medium (e.g., 256 levels)		Slight banding
High (e.g. 14-bit, 16384 levels)		Smooth gradient

6 Method to Solve it (Bit Depth)



1. **Increase Bit Depth (A/D Resolution)**
   - Use a sensor/converter with higher bit depth (e.g., from 10-bit to 14-bit, or 16-bit).
   - More levels → finer steps → seamless gradients.



2. **Use Dithering (Process). (Replaces sampling rate grid).**
   - Apply randomized noise during quantization to 'break up' visible banding, perceptually masking.



3. **Utilize Advanced Post-Processing (Software).**
   - Employ color space conversions, deeper bit-depth image formats (e.g., TIFF), and gradient smoothing algorithms (improves appearance).

7 Optimized Image with Seamless Gradient.



- When the bit depth is high (more quantization levels), the image has seamless, natural gradients.
- The gradient steps are too small to be perceived.
- Gradual tone changes are preserved.

8 Conclusion.

- Improper quantization (low bit depth) captures too few distinct digital values, leading to visible banding (posterization) in smooth gradients.
- Increasing the quantization levels (higher bit depth) or utilizing dithering can provide more steps for a seamless gradient and eliminate contouring artifacts.
- Finer steps are key to preserving gradients.

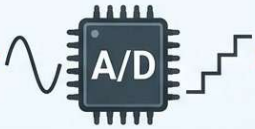
33 A low-bit image loses important details. Explain how quantization affects information preservation. Maximize detail capture by understanding bit depth and tonal range.

1 Full-Bit Scene (Original object)



- This is a real-world complex scene with a wide range of continuous tonal values (e.g., skin, clothing).
- It has thousands of possible shades and subtle gradients.
- We want to capture this high-fidelity information digitally.

2 The A/D Converter (Digitization)



- The analog signal is converted to digital steps.
- The number of bits (bit depth) determines the number of "steps" or discrete levels available to represent the original continuous values.

3 Signal Quantization (Low Bit Depth)



- Continuous ranges of the input signal are rounded (quantized) to the nearest available discrete level.
- A low bit depth (few levels) means large continuous gradient sections are mapped to single "bins".
- Fine details (smooth tonal shifts) are lost forever.

4 The Low-Bit Image (Artifacts)



- Because of few bits (fewer color levels), the image has severe contouring and banding.
- Subtle gradients appear as harsh steps (lost information).
- Perceptible visual noise and artifacts (e.g., "crushed" blacks, "blown out" highlights) are visible.
- Important textural details are merged into single colors.

5 Effect of Bit Depth on Information

	Bit Depth	Info Quality
	**Low** (e.g., 2-bit or 4-bit)	**Failure** (Contouring, lost details)
	**Medium** (e.g., 8-bit or 10-bit)	**Average** (Visible gradients, acceptable details)
	**High** (e.g., 16-bit)	**Precise** (Rich tonal range, preserved details)

6 Method to Preserve Details



1. Increase Sensor Bit Depth
- Use a sensor with a higher-resolution A/D converter (e.g., from 10-bit to 16-bit raw capture).

2. Apply Advanced Dithering
- Use software or hardware dithering before quantization to break up banding with randomized noise, perceptually hiding the lost detail levels.

3. Optimize Tone Curve
- Use software to remap the available bits to maximize the detail in critical areas (e.g., shadows or highlights) before final encoding.

7 Optimized/High-Bit Image



- When the bit depth is high (more levels), tonal details are preserved and precise.
- Subtle gradients and skin textures are smooth and continuous.
- Full information fidelity is achieved.

8 Conclusion

- Bit depth determines the available levels to represent analog information. A low bit depth (low-bit image) causes quantization to "crush" tonal values, merging continuous ranges into blocky discrete levels and losing essential through a reduction in tdelity. Information lost at this stage cannot be fully recovered.
- Utilizing higher-bit sensors and applying advanced software can improve the perceived quality, but adequate initial capture is fundamental.
- Proper bit depth is key to preserving complex information details.

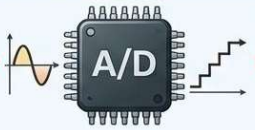
40. A digital image shows quantization noise. Explain its origin and suggest mitigation techniques. Understand the visual artifacts of limited bit depth and how to control them.

1 Analog Input Signal (Scene)



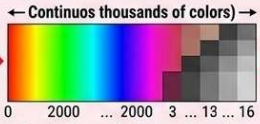
- This is a perfect, smooth analog signal (or scene).
- It has infinite color and tonal values.
- We want to digitize this scene without artifacts.

2 The A/D Converter (ADC)



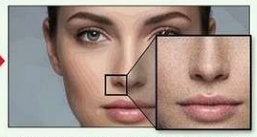
- The ADC samples the signal at continuous points.
- It measures the signal intensity (voltage) over a range.
- A crucial step: it must round (quantize) values.

3 Signal Quantization (Origin)



- The ADC has a limited number of "buckets" (bit depth).
- Continuous input values are rounded (quantized) to the nearest discrete level.
- Large range values merge into single "bins".
- The error (difference) is quantization noise.

4 The Quantized (Dithered) Result.



- Severe banding makes gradients look stepping.
- Perceptual noise (artifacts) are visible.
- Fine details (smooth shifts) are lost.
- Random dithering is applied to perceptually ***improve*** but it ***increases*** noise floor.

5 Effect of Bit Depth on Quality

Bit Depth	Gradient Example	Visual Quality
Low (e.g., 2-bit)		Banding/ Contouring (lost levels)
Medium (e.g., 8-bit)		Average (standard levels)
High (e.g., 16-bit)		**Smooth** (clear details)

6 Mitigation Techniques



1. Increase Bit Depth (A/D Resolution)
- Use a higher-resolution ADC. Example: from 8-bit to 12-bit, 14-bit, or higher.

2. Apply Advanced Dithering
- Add low-level, random noise *before* quantization. This converts noticeable banding into subtle, preferable textural noise.

3. Post-Processing & Filtering (Software)
- Apply noise-reduction algorithms or smoothing filters to reduce perceptible noise and correct artifacts (like banding/halos).

7 Mitigated/Improved Image



- Smooth gradients are preserved.
- Tonal levels are vast and detailed.
- Details are preserved.
- Banding artifacts are eliminated.
- The noise texture is natural (from better capture and dithering).

8 Conclusion

- Quantization noise originates from the limited A/D bit depth, causing the rounding of infinite tonal values into discrete buckets, leading to artifacts like contouring.
- Increasing the bit depth (ADC resolution) is the primary correction, providing more buckets for a precise signal.
- Advanced dithering can be used *perceptually* to mask banding but adds noise.
- Post-processing can mitigate some noise afterwards.
- Bit depth is fundamental to digital image fidelity.

43. In a facial recognition system, explain how resolution impacts feature extraction accuracy.

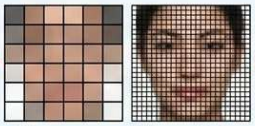
Ensure optimal capture for deep learning by understanding resolution and features.

1 Original Scene with Face.



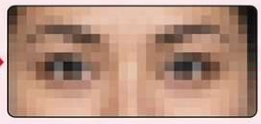
- This is a real-world clear face.
- It has complex details like eye-corners, nose shape, and mouth curvature.
- We want to capture this using a digital sensor.

2 The Problem with Low Sampling.



- Using a sensor with large pixels (low resolution).
- Small features like eyelashes and iris details cannot be resolved.
- The face becomes a coarse grid pattern.
- Neighboring face details merge.

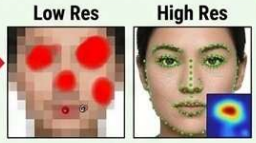
3 Sampled Feature Map (Low Res).



- Few pixels define key features.
- Feature boundaries (e.g., edge of eye, nasolabial fold) are indistinct.
- Specific feature points cannot be accurately located.
- Feature descriptors (like iris pattern or skin texture) are lost.

4 The Problem for Feature Extractor.

Low Res | High Res



- Undersampled features result in feature extraction failure.
- Extractor cannot reliably locate key facial landmarks.
- Correct resolution allows precise extraction of many feature points.

5 Effect of Resolution on Feature Map.

	Resolution	Feature Map Quality
	Coarse (Very few pixels/face)	Failure (No discernible features)
	Medium (Adequate pixels/face)	Average (Some features discernible but points have low confidence)
	High (Detailed features)	Precise (Many accurate feature landmarks extracted)

6 Method to Improve Extraction Accuracy.



1. Increase Sensor Resolution.
- Use a higher-resolution sensor with smaller pixels to sample the face more finely.



2. Increase Resolution of Target.
- Use optical zoom or frame the shot closer to fill more of the existing sensor with the face. This effectively samples the face with more pixels.



3. Use Super-Resolution Processing (Software).
- If a higher resolution capture le impossible, use advanced software (like deep-learning based super-resolution) to reconstruct high-frequency details. This is not a substitute for better capture but can help in niche cases.

7 Optimized Feature Map (High Res)



- When the sampling rate/resolution is high (more pixels), the face features are sharp.
- Detailed face landmark points can be precisely located and defined.
- A high-confidence facial descriptor is generated.

8 Conclusion

- In facial recognition, resolution is critical for feature extraction.
- Undersampling (low resolution) leads to feature loss and extraction failure, as complex details are missed.
- Increasing the sampling rate (higher sensor resolution or closer shot) directly improves extraction accuracy by providing more pixels to define facial features.
- Using advanced software can be a helpful, but secondary, method.
- Ensure robust feature capture through optimal visual design.

46. A system requires detecting small objects in images. Explain how sampling and resolution should be adjusted.

Optimize your imaging system for small object detection using proper sampling and high resolution.

1 Original Scene with Small Objects



- This is a real-world complex scene with small objects (a distant person and a small sign).
- They occupy very few pixels in a typical sensor view.

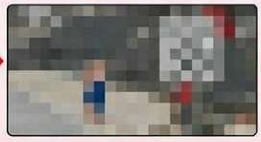
2 The Problem with Low Resolution



Using a sensor with large pixels or low total pixel count.

- Fewer pixels (low sampling rate).
- The small objects are not sampled with enough pixels to be resolved.
- The small sign and person become unidentifiable blobs.
- Neighboring pixels blend over the small object details.

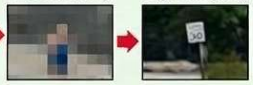
3 Sampled Image (Low Sampling Rate)



- Small objects are poorly sampled.
- Small details merge into single pixels.
- The objects are 'undersampled' and invisible to a standard detector.
- No usable features remain.

4 Adjustments for Improvement

Low Sampling | High Sampling



- **Option A:** Increase the sampling rate (Use more smaller pixels).
- **Option B:** Increase the resolution of the ***object's image*** (Use optical zoom to fill more of the existing sensor with the object).
- **Option C:** Combine both for best results.

5 Effect of Resolution on Detection

	Resolution	Detection
	**Low** (Fewer pixels/obj)	**Fails** (Small object not found)
	**Medium** (Adequate pixels/obj)	**Average** (Found, but low confidence)
	**High** (Many pixels/obj)	**Succeeds** (Object found, type identified)

6 Correction Methods (Step-by-Step)



Method 1 — **Increase System Resolution**
Use a higher-resolution sensor with smaller, more numerous pixsls (e.g., from 2MP to 20MP sensor).



Method 2 — **Adjust Sampling Region**
Use a longer focal length (telephoto zoom) or move the camera closer to the object to occupy more of the sensor. This effectively samples the small object with more pixels.



Method 3 — **Use Super-Resolution Software**
Apply advanced image processing algorithms like deep-learning based super-resolution to synthetically increase resolution (reconstructs finer details). This is an adjunct, not a substitute for better capture.

7 Corrected Image with High Resolution



- When the sampling rate/resolution on the object is high, details are preserved.
- The object is properly sampled.
- Detectors can successfully find and classify the small objects.

8 Conclusion

- To detect small objects, the system requires a high sampling rate on the object itself. Small objects are easily lost with improper sampling (undersampling).
- Increasing sensor resolution or using optical zoom ensures enough pixels are allocated to the small object.
- Proper resolution is key to preserving essential object details for reliable detection and classification.
- A combination of higher sensor resolution and optical zoom is most effective.
- Super-resolution software can be a helpful final step.

Code of Conduct:

I RAHIMA BANU N-192525009 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: RAHIMA BANU N